Other-Directed Emotion Regulation and Sensitivity to Fairness

Contents

Loading data 1

Sensitivity to fairness calculation 3

Dataframe for analyses... 4

Behavioral Analyses 5
Acceptance Rate 5
Anterior Insula Categorical Plot 8

Activation 11
dlPFC, S > NS behavioral sensitivity, Extrinsic Affect Improving (Other-Directed Affect Improving) 11

PPI 13
Amygdala, S > NS behavioral sensitivity 13
Amygdala-dmPFC, S > NS behavioral sensitivity, Extrinsic Worsening (Other-Directed Affect Worsening) 15

Behavioral Plots (supplement) 18

Loading data

Loading in the .csv files with participant demographic, behavioral (e.g., EROS scores), and neuroimaging data (e.g., extracted cluster values). If you are interested in the data, please email.

```r
# Load .csvs with subject data
full_df <- read_csv("/Users/tur61139/Documents/GitHub/rf1-kos/df_MRdata_n138.csv")
```

```
## Rows: 12494 Columns: 81
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (7): race, ethnicity, sub_gender, gender_at_birth, endowment_label, co...
## dbl  (73): sub_id, tppid, sub_age, nTrial, run, cue_Onset, event_duration, b...
## date  (1): sub_dob
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
df <- read_csv("/Users/tur61139/Documents/GitHub/rf1-kos/df_MRcovar_n138.csv")
```

```
## Rows: 138 Columns: 28
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (28): sub_id, gender_female, gender_male, sub_age_MC, avg_fdmean_MC, avg...
##
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
neural_df <- read_csv("/Users/tur61139/Documents/GitHub/rf1-norms-hpc/n138_filterfunc.csv")
```

```
## Rows: 138 Columns: 13
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (13): sub_id, AI_act_C11, AI_act_C12, dACC_act_C11, dACC_act_C12, mPFC_a...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
df_extracted <- read_csv("/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/compiled_
```

```
## Rows: 138 Columns: 45
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (45): sub_id, act_AlphaDiff-ExtBet-Int_C12_zstat3_cluster1, act_AlphaDif...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
full_df <- left_join(full_df, df, by = "sub_id")
full_df <- full_df |>
  rename(avg_ext_bet = avg_ext_bet.x,
         avg_ext_wor = avg_ext_wor.x,
         avg_int_bet = avg_int_bet.x,
         avg_int_wor = avg_int_wor.x,
         reldep = reldep.x,
         gender_female = gender_female.x,
         gender_male = gender_male.x)

# Trial-level behavioral accept/reject variables
full_df <- full_df |>
  mutate(offer = R_Option + L_Option,
         beh_fairness = offer / Endowment,
         accept = ifelse(accept_reject == "Accept", 1, 0),

         # Percentage label for plotting
         beh_fairness_perc = case_when(beh_fairness >= 0.060 & beh_fairness <= 0.0625 ~ "5%",
                                       beh_fairness >= 0.09375 & beh_fairness <= 0.125 ~ "10%",
                                       abs(beh_fairness - 0.25) < 1e-6 ~ "25%",
                                       abs(beh_fairness - 0.50) < 1e-6 ~ "50%",
                                       TRUE ~ NA_character_),
         # Numeric binned version for modeling (since )
         beh_fairness_num = case_when(beh_fairness_perc == "5%" ~ 0.05,
                                      beh_fairness_perc == "10%" ~ 0.10,
                                      beh_fairness_perc == "25%" ~ 0.25,
                                      beh_fairness_perc == "50%" ~ 0.50,
                                      TRUE ~ NA_real_),
         # Factor version for plotting
         beh_fairness_perc = factor(beh_fairness_perc, levels = c("5%", "10%", "25%", "50%")))
```

Sensitivity to fairness calculation

Calculating the 4 different slopes for sensitivities to fairness based on the behavioral data: all-trial, social trials only, nonsocial trials only, then the difference scored social - nonsocial.

```
## Sensitivity to fairness
# Function to extract subject-level fairness slope
get_fairness_slope <- function(df, label) {
  # Fixed effect of behavioral fairness, random effect with subject
  model <- glmer(accept ~ beh_fairness + (1 + beh_fairness | sub_id), data = df, family = binomial,
    control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000)))
  fe <- fixef(model)[["beh_fairness"]]
  re <- ranef(model)$sub_id[, "beh_fairness"]
  subjIDs <- as.numeric(rownames(ranef(model)$sub_id))
  slopes <- fe + re

  tibble(sub_id = subjIDs, !!label := slopes)
}

# Estimate fairness slopes across all trials, social trials, and nonsocial trials
slope_all_df <- get_fairness_slope(full_df, "slope_all")
slope_social_df <- get_fairness_slope(filter(full_df, condition == "social"), "slope_social")
slope_nonsocial_df <- get_fairness_slope(filter(full_df, condition == "nonsocial"), "slope_nonsocial")

# Merge slopes into one df and compute social - nonsocial difference score
slope_df_wide <- reduce(list(slope_all_df, slope_social_df, slope_nonsocial_df), full_join, by = "sub_i
  mutate(diff_slope = slope_social - slope_nonsocial)
# Printing first 15 subs
print(head(slope_df_wide, 15), digits=3)
```

```
## # A tibble: 15 x 5
##    sub_id slope_all slope_social slope_nonsocial diff_slope
##     <dbl>     <dbl>        <dbl>           <dbl>      <dbl>
##  1  10369      16.1        16.6             14.0      2.61
##  2  10402      11.9        15.7             10.6      5.04
##  3  10478      35.5        30.2             40.8    -10.5
##  4  10486      10.8         6.01            13.9     -7.89
##  5  10559      22.3        35.8             15.7     20.1
##  6  10581      11.3        15.9             10.4      5.55
##  7  10584      33.7        36.3             30.2      6.13
##  8  10589      22.1        29.2             17.6     11.6
##  9  10596      22.1        31.7             19.4     12.4
## 10  10606      31.0        31.5             32.0     -0.481
## 11  10608      16.2        13.1             30.3    -17.2
## 12  10636      32.9        35.2             29.4      5.78
## 13  10640      41.6        40.8             38.2      2.62
## 14  10641      30.2        29.4             28.7      0.728
## 15  10642      15.1        13.5             18.7     -5.21
```

```
# Add slope variables back into the trial-level dataframe
full_df <- left_join(full_df, slope_df_wide, by = "sub_id")
```

Dataframe for analyses...

```
# Change the names so that it's more legible for figures
pretty_names <- c(gender_female = "Female", gender_male = "Male", sub_age = "Age", sub_gender = "Gender
                  usi_total = "USI Score", ADI_NATRANK = "ADI (National Rank)", reldep = "Relative Depri
                  avg_ext_bet.x = "Extrinsic Improving",
                  avg_ext_wor.x = "Extrinsic Worsening",
                  avg_int_bet.x = "Intrinsic Improving",
                  avg_int_wor.x = "Intrinsic Worsening",
                  slope_all = "Sensitivity to Fairness",
                  diff_slope = "Sensitivity to Fairness, S-NS",
                  slope_social = "Social Sensitivity to Fairness",
                  slope_nonsocial = "Nonsocial Sensitivity to Fairness")

# Subject-level covariates pulled from full_df
subject_vars <- full_df |>
  select(sub_id, starts_with("avg_"), ends_with("_MC")) |>
  distinct()

# Add subject-level covariates to df used for downstream analyses
df <- left_join(df, subject_vars, by = "sub_id")

# Long-format slope dataframe for social vs. nonsocial plots
slope_df <- slope_df_wide |>
  left_join(full_df |>
              select(sub_id, avg_ext_bet, avg_ext_wor, avg_int_bet, avg_int_wor, reldep) |>
              distinct(), by = "sub_id") |>
  pivot_longer(cols = c(slope_social, slope_nonsocial), names_to = "condition", values_to = "slope") |>
  mutate(condition = ifelse(condition == "slope_social", "social", "nonsocial"))

# Subject-level acceptance rates within each condition
accept_rates <- full_df |>
  group_by(sub_id, condition) |>
  summarise(accept_rate = mean(accept, na.rm = TRUE), .groups = "drop") |>
  left_join(full_df |>
              select(sub_id, avg_ext_bet, avg_ext_wor, avg_int_bet, avg_int_wor, reldep) |>
              distinct(), by = "sub_id")

# Final df for modeling
df <- left_join(df_extracted, df, by = "sub_id")

# Create df with subject demographic and questionnaire information from long-format trial df
demog_df <- full_df |>
  group_by(sub_id) |>
  slice(1) |>
  ungroup() |>
  select(sub_id, gender_female ,gender_male, sub_age, race, ethnicity, sub_gender,
         usi_total, ADI_NATRANK, reldep, # USI and ADI and relative deprivation
         avg_ext_bet, avg_ext_wor, avg_int_bet, avg_int_wor, # EROS scores
         slope_all, diff_slope, slope_social, slope_nonsocial)
```

Behavioral Analyses

Acceptance Rate

Looking at the acceptance rate for offers across the 4 levels of fairness. In particular, want to see if there is a significant difference in 1) fairness on offer acceptance and 2) fairness and social context (human vs. computer partner) on offer acceptance.

```r
fairness <- glmer(accept ~ beh_fairness_perc + (1 | sub_id), data = full_df, family = binomial)
summary(fairness)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_perc + (1 | sub_id)
##    Data: full_df
##
##      AIC      BIC   logLik deviance df.resid
##     7518     7555    -3754     7508    12489
##
## Scaled residuals:
##    Min     1Q Median     3Q    Max
## -96.29  -0.27   0.06   0.25   7.71
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  sub_id (Intercept) 7.51     2.74
## Number of obs: 12494, groups:  sub_id, 138
##
## Fixed effects:
##                      Estimate Std. Error z value             Pr(>|z|)
## (Intercept)           -1.0646     0.2436   -4.37             0.000012 ***
## beh_fairness_perc10%   0.7378     0.0809    9.12 < 0.0000000000000002 ***
## beh_fairness_perc25%   3.9793     0.0962   41.37 < 0.0000000000000002 ***
## beh_fairness_perc50%   6.0291     0.1348   44.73 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) b__10% b__25%
## bh_frns_10% -0.180
## bh_frns_25% -0.184  0.527
## bh_frns_50% -0.139  0.379  0.531
```

```r
car::Anova(fairness, type = 3)
```

```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: accept
##                    Chisq Df           Pr(>Chisq)
## (Intercept)         19.1  1             0.000012 ***
## beh_fairness_perc 2786.0  3 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
fairness_condition <- glmer(accept ~ beh_fairness_perc * condition + (1 | sub_id), data = full_df, famil
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge with max|grad| = 0.00511866 (tol = 0.002, component 1)
```

```r
summary(fairness_condition)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: accept ~ beh_fairness_perc * condition + (1 | sub_id)
##    Data: full_df
##
##      AIC      BIC   logLik deviance df.resid
##     7521     7588    -3752     7503    12485
##
## Scaled residuals:
##    Min     1Q Median     3Q    Max
## -88.99  -0.28   0.06   0.26   7.62
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  sub_id (Intercept) 7.52     2.74
## Number of obs: 12494, groups:  sub_id, 138
##
## Fixed effects:
##                                     Estimate Std. Error z value
## (Intercept)                          -1.0197     0.2504   -4.07
## beh_fairness_perc10%                  0.7206     0.1127    6.39
## beh_fairness_perc25%                  3.9975     0.1258   31.79
## beh_fairness_perc50%                  6.1680     0.1856   33.24
## conditionsocial                      -0.0911     0.1183   -0.77
## beh_fairness_perc10%:conditionsocial  0.0356     0.1589    0.22
## beh_fairness_perc25%:conditionsocial -0.0315     0.1614   -0.20
## beh_fairness_perc50%:conditionsocial -0.2517     0.2326   -1.08
##                                                Pr(>|z|)
## (Intercept)                          0.00004659644 ***
## beh_fairness_perc10%                 0.00000000016 ***
## beh_fairness_perc25%                 < 0.0000000000000002 ***
## beh_fairness_perc50%                 < 0.0000000000000002 ***
## conditionsocial                                    0.44
## beh_fairness_perc10%:conditionsocial               0.82
## beh_fairness_perc25%:conditionsocial               0.85
## beh_fairness_perc50%:conditionsocial               0.28
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) bh__10% bh__25% bh__50% cndtns b__10%: b__25%:
## bh_frns_10% -0.246
## bh_frns_25% -0.243  0.527
## bh_frns_50% -0.171  0.359   0.441
## conditinscl -0.233  0.514   0.458   0.311
## bh_frn_10%:  0.174 -0.697  -0.344  -0.233  -0.744
```

```
## bh_frn_25%:  0.171 -0.377  -0.644  -0.231  -0.733  0.545
## bh_frn_50%:  0.119 -0.262  -0.235  -0.686  -0.509  0.379   0.373
## optimizer (Nelder_Mead) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.00511866 (tol = 0.002, component 1)
```

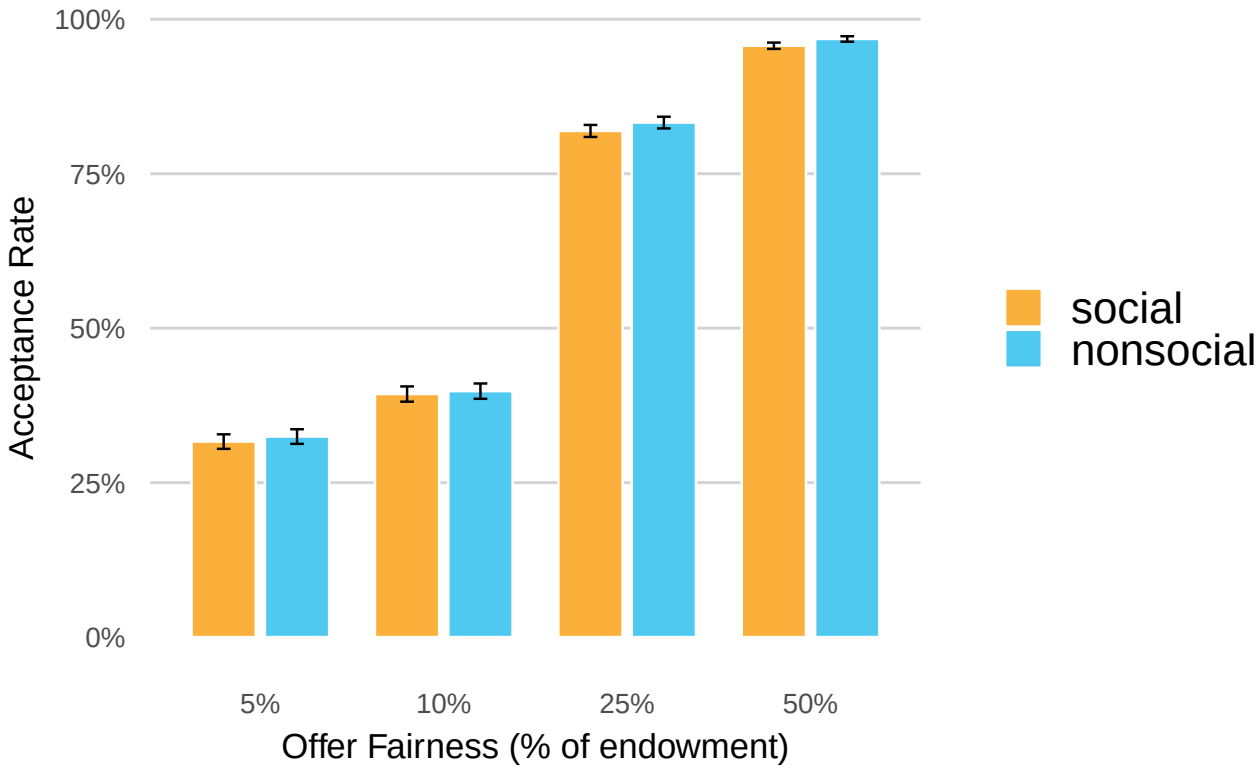
```r
car::Anova(fairness_condition, type = 3)
```

```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: accept
##                                Chisq Df           Pr(>Chisq)
## (Intercept)                    16.58  1             0.000047 ***
## beh_fairness_perc            1722.74  3 < 0.0000000000000002 ***
## condition                       0.59  1                 0.44
## beh_fairness_perc:condition     1.65  3                 0.65
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Summarise with SE and put "social" first
plot_df <- full_df |>
  group_by(beh_fairness_perc, condition) |>
  summarise(n = dplyr::n(),
            accept_rate = mean(accept_binary, na.rm = TRUE),
            sd_accept = sd(accept_binary, na.rm = TRUE),
            se_accept = sd_accept / sqrt(n), .groups = "drop") |>
  mutate(condition = factor(condition, levels = c("social", "nonsocial")),
         ymin = pmax(0, accept_rate - se_accept),
         ymax = pmin(1, accept_rate + se_accept))

pd <- position_dodge(width = 0.8)  # shared dodge object

ggplot(plot_df, aes(x = beh_fairness_perc, y = accept_rate, fill = condition)) +
  geom_hline(yintercept = c(0.25, 0.50, 0.75, 1.00),
           linetype = "solid",
           color = "lightgray",
           linewidth = 0.6) +
  geom_col(width = 0.7, color = "white", position = pd) +
  geom_errorbar(aes(ymin = ymin, ymax = ymax), width = 0.15, position = pd, linewidth = 0.5) +
  scale_y_continuous(name = "Acceptance Rate",
                     breaks = c(0, 0.25, 0.50, 0.75, 1.00),
                     labels = scales::label_percent(accuracy = 1)) +
  coord_cartesian(ylim = c(0, 1)) +
  scale_x_discrete(name = "Offer Fairness (% of endowment)") +
  scale_fill_manual(values = c("social" = "#FBB03B", "nonsocial" = "#50C9F1"), breaks = c("social", "no
  theme_minimal(base_size = 24) +
  theme(panel.grid = element_blank(), axis.title = element_text(size = 15), axis.text = element_text(si
```



Significant for offer fairness, but not the interaction term.

Anterior Insula Categorical Plot

Using data from the categorical model in FSL (level 1 = 5% offer, and so on) to demonstrate the anterior insula tracking offer fairness regardless of a low or high endowment amount.

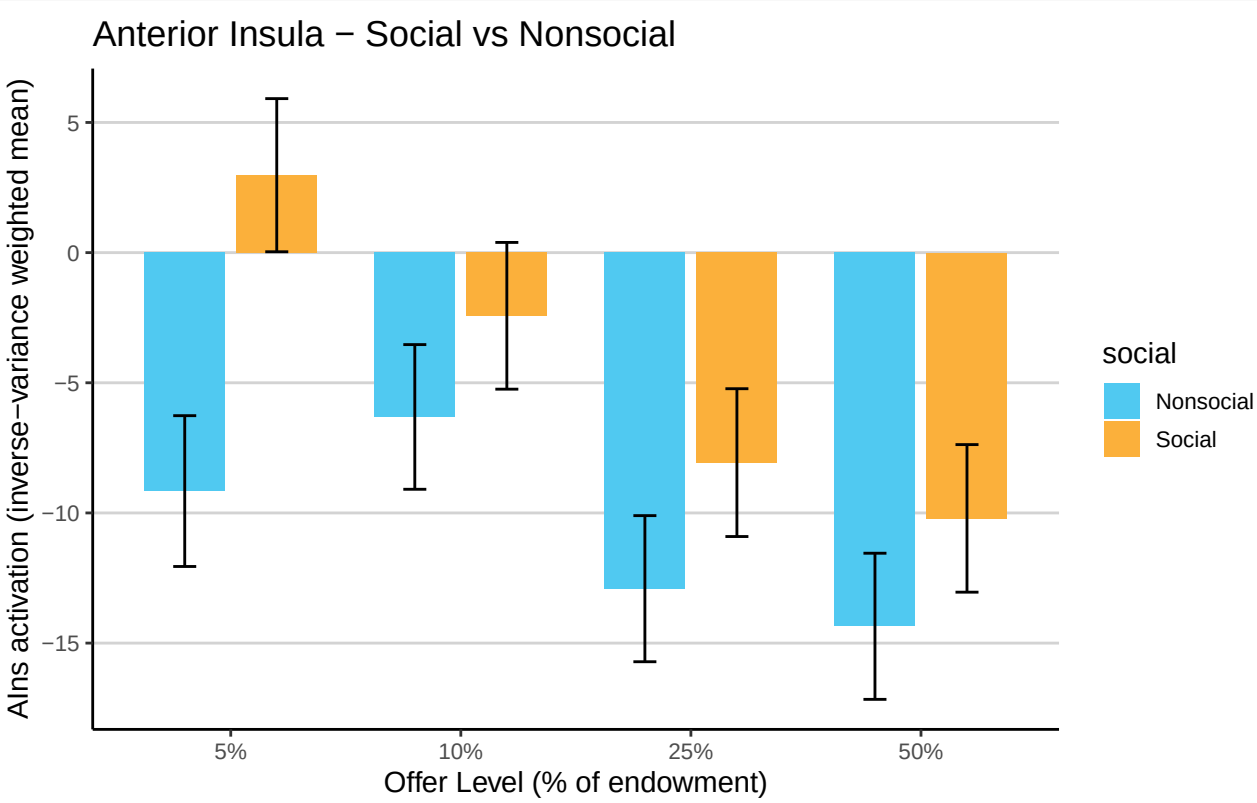
```
# 8 conditions
conditions <- c("/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty
  "/Users/tur61139/Documents/GitHub/rf1-norms/derivatives/imaging_plots/thr_seed-AIns.nii.gz_model-3b_ty

# Weighted means and SEM function thx chat
summarise_condition <- function(cond) {
  val_file <- paste0(cond, ".txt")
  var_file <- file.path(dirname(cond), paste0("var_", basename(cond), ".txt"))
  vals <- scan(val_file, what = numeric(), quiet = TRUE)
  vars <- scan(var_file, what = numeric(), quiet = TRUE)
  if (length(vals) != length(vars)) stop("Lengths differ for ", cond)
  w <- 1 / vars
  tibble(
    condition = basename(cond),
    mean = sum(w * vals) / sum(w),
    sem  = sqrt(1 / sum(w)),
    n    = length(vals))
}
```

```r
# Weighted means and SEMs
summary_df <- map_dfr(conditions, summarise_condition)

# Parse social and level
cat_plot_df <- summary_df |>
  mutate(social = if_else(str_detect(condition, "nonsocial"), "Nonsocial", "Social"),
    level_digit = str_match(condition, "level(\\d)")[,2],
    level = factor(level_digit,
                   levels = c("1","2","3","4"),
                   labels = c("5%","10%","25%","50%")),
    cope_num = as.integer(str_match(condition, "cope-(\\d+)")[,2]))

# Plot
ggplot(cat_plot_df, aes(x = level, y = mean, fill = social)) +
  geom_hline(yintercept = c(-15, -10, -5, 0, 5),
           linetype = "solid",
           color = "lightgray",
           linewidth = 0.6) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_errorbar(aes(ymin = mean - sem,
                    ymax = mean + sem),
                position = position_dodge(width = 0.8), width = 0.2, linewidth=0.6) +
  scale_fill_manual(values = c("Social" = "#FBB03B", "Nonsocial" = "#50C9F1")) +
  labs(x = "Offer Level (% of endowment)",
       y = "AIns activation (inverse-variance weighted mean)",
       title = "Anterior Insula - Social vs Nonsocial") +
  theme_classic(base_size = 13)
```



```r
# Subject level df for stats
cat_stats_df <- map_dfr(conditions, function(cond) {
  val_file <- paste0(cond, ".txt")
  vals <- scan(val_file, what = numeric(), quiet = TRUE)
  tibble(sub = factor(seq_along(vals)), condition = basename(cond), value = vals)}) |>
  mutate(social = if_else(str_detect(condition, "nonsocial"), "Nonsocial", "Social"),
         level_digit = str_match(condition, "level(\\d)")[, 2],
         level = factor(level_digit, levels = c("1", "2", "3", "4"),
                        labels = c("5%", "10%", "25%", "50%")))

# Mixed model for fairness level, sociality, and interaction
m_cat <- lmer(value ~ social * level + (1 | sub), data = cat_stats_df)
anova(m_cat)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF DenDF F value       Pr(>F)
## social        12798   12798     1   959   33.22 0.0000000111 ***
## level         17276    5759     3   959   14.95 0.0000000016 ***
## social:level   3253    1084     3   959    2.81        0.038 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m_cat)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ social * level + (1 | sub)
##    Data: cat_stats_df
##
## REML criterion at convergence: 10084
##
## Scaled residuals:
##    Min     1Q Median     3Q    Max
## -5.622 -0.536  0.013  0.539  3.840
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  sub      (Intercept) 901      30.0
##  Residual             385      19.6
## Number of obs: 1104, groups:  sub, 138
##
## Fixed effects:
##                       Estimate Std. Error      df t value    Pr(>|t|)
## (Intercept)            -10.075      3.054 247.081   -3.30      0.0011 **
## socialSocial            12.738      2.363 959.000    5.39 0.000000088 ***
## level10%                 0.986      2.363 959.000    0.42      0.6765
## level25%                -4.560      2.363 959.000   -1.93      0.0539 .
## level50%                -5.977      2.363 959.000   -2.53      0.0116 *
## socialSocial:level10%   -8.321      3.342 959.000   -2.49      0.0129 *
## socialSocial:level25%   -7.801      3.342 959.000   -2.33      0.0198 *
## socialSocial:level50%   -7.592      3.342 959.000   -2.27      0.0233 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Correlation of Fixed Effects:
##             (Intr) sclScl lvl10% lvl25% lvl50% sS:10% sS:25%
## socialSocil -0.387
## level10%    -0.387  0.500
## level25%    -0.387  0.500  0.500
## level50%    -0.387  0.500  0.500  0.500
## sclScl:l10%  0.274 -0.707 -0.707 -0.354 -0.354
## sclScl:l25%  0.274 -0.707 -0.354 -0.707 -0.354  0.500
## sclScl:l50%  0.274 -0.707 -0.354 -0.354 -0.707  0.500  0.500
```

Activation

dlPFC, S > NS behavioral sensitivity, Extrinsic Affect Improving (Other-Directed Affect Improving)

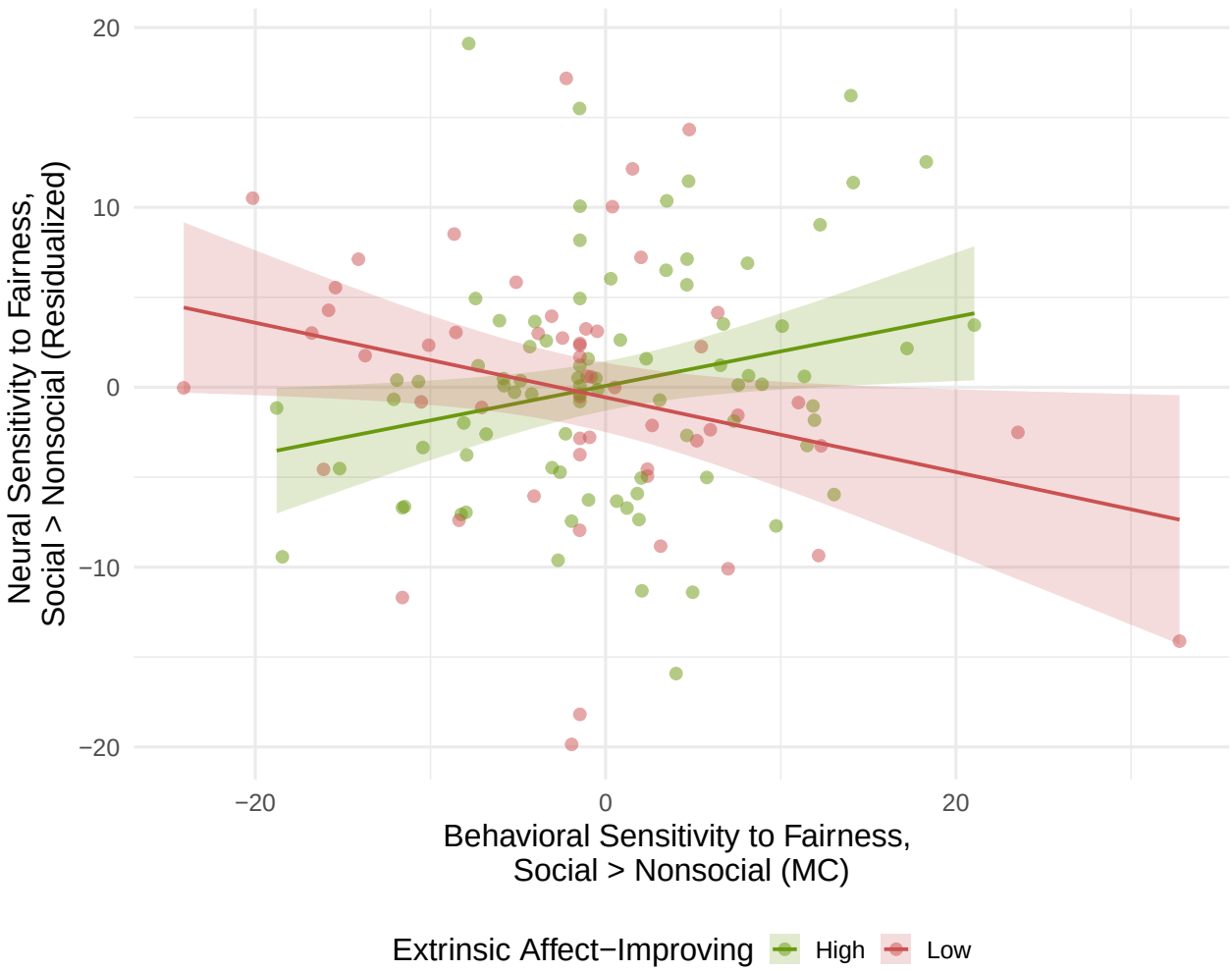
```r
# Create High/Low grouping by median split
df$ext_bet_group <- ifelse(subject_vars$avg_ext_bet_MC.x >= median(subject_vars$avg_ext_bet_MC.x, na.rm
                           "High", "Low")

# Get the full 12-color color-blind friendly palette
cb_palette <- SteppedSequential5Steps

# Choose colors
cb_pair <- cb_palette[c(3, 11)]
names(cb_pair) <- c("Low", "High")


# Regression scatterplot with colorblind-safe palette and residualized X and Y
ggplot(df, aes(
  x = diff_slope,
  y = residuals(lm(`act_AlphaDiff-ExtBet-Int_C12_zstat3_cluster1` ~ sub_age_MC + gender_female + gender_
  color = ext_bet_group,
  fill = ext_bet_group)) +
  geom_point(alpha = 0.5, size = 3) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.2) +
  scale_color_manual(values = cb_pair) +
  scale_fill_manual(values = cb_pair) +
  labs(
    x = "Behavioral Sensitivity to Fairness, \nSocial > Nonsocial (MC)",
    y = "Neural Sensitivity to Fairness, \nSocial > Nonsocial (Residualized)",
    color = "Extrinsic Affect-Improving",
    fill = "Extrinsic Affect-Improving") +
  theme_minimal(base_size = 18) +
  theme(legend.position="bottom")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



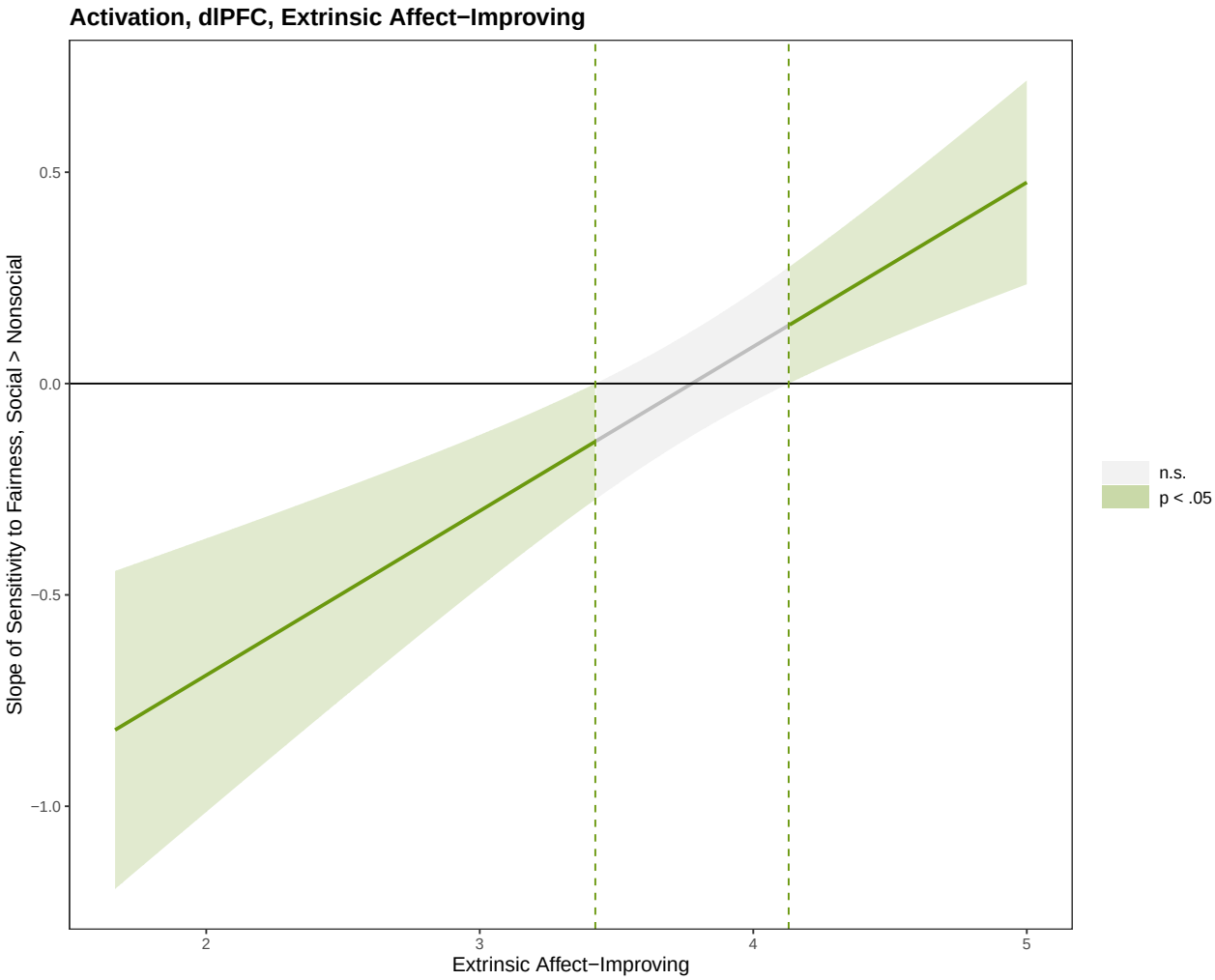

```
# Johnson-Neyman plot version
m_jn <- lm(`act_AlphaDiff-ExtBet-Int_C12_zstat3_cluster1` ~ diff_slope * avg_ext_bet.x +  sub_age_MC + g
  data = df)

# Obs range of moderator
mf <- model.frame(m_jn)
mod_rng <- range(mf$avg_ext_bet.x, na.rm = TRUE)

# Johnson-Neyman analysis
jn <- johnson_neyman(m_jn,
  pred = "diff_slope",
  modx = "avg_ext_bet.x",
  alpha = 0.05,
  mod.range = mod_rng,
  sig.color = cb_palette[11],
  insig.color = "gray",
  line.thickness = 1,
  y.label = "Slope of Sensitivity to Fairness, Social > Nonsocial",
  modx.label = "Extrinsic Affect-Improving",
  title = "Activation, dlPFC, Extrinsic Affect-Improving")
jn
```

```
## JOHNSON-NEYMAN INTERVAL
```

```
##
## When avg_ext_bet.x is OUTSIDE the interval [3.42, 4.13], the slope of
## diff_slope is p < .05.
##
## Note: The range of observed values of avg_ext_bet.x is [1.67, 5.00]
```



PPI

Amygdala, S > NS behavioral sensitivity

```
diff_amyg <- lm(`amyg_AlphaDiff_C12` ~ sub_age_MC + gender_female + gender_male + avg_fdmean_MC + avg_ts
# NO BEHAV COVAR OF INTEREST
summary(diff_amyg)
```

```
##
## Call:
## lm(formula = amyg_AlphaDiff_C12 ~ sub_age_MC + gender_female +
##     gender_male + avg_fdmean_MC + avg_tsnr_MC, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
```

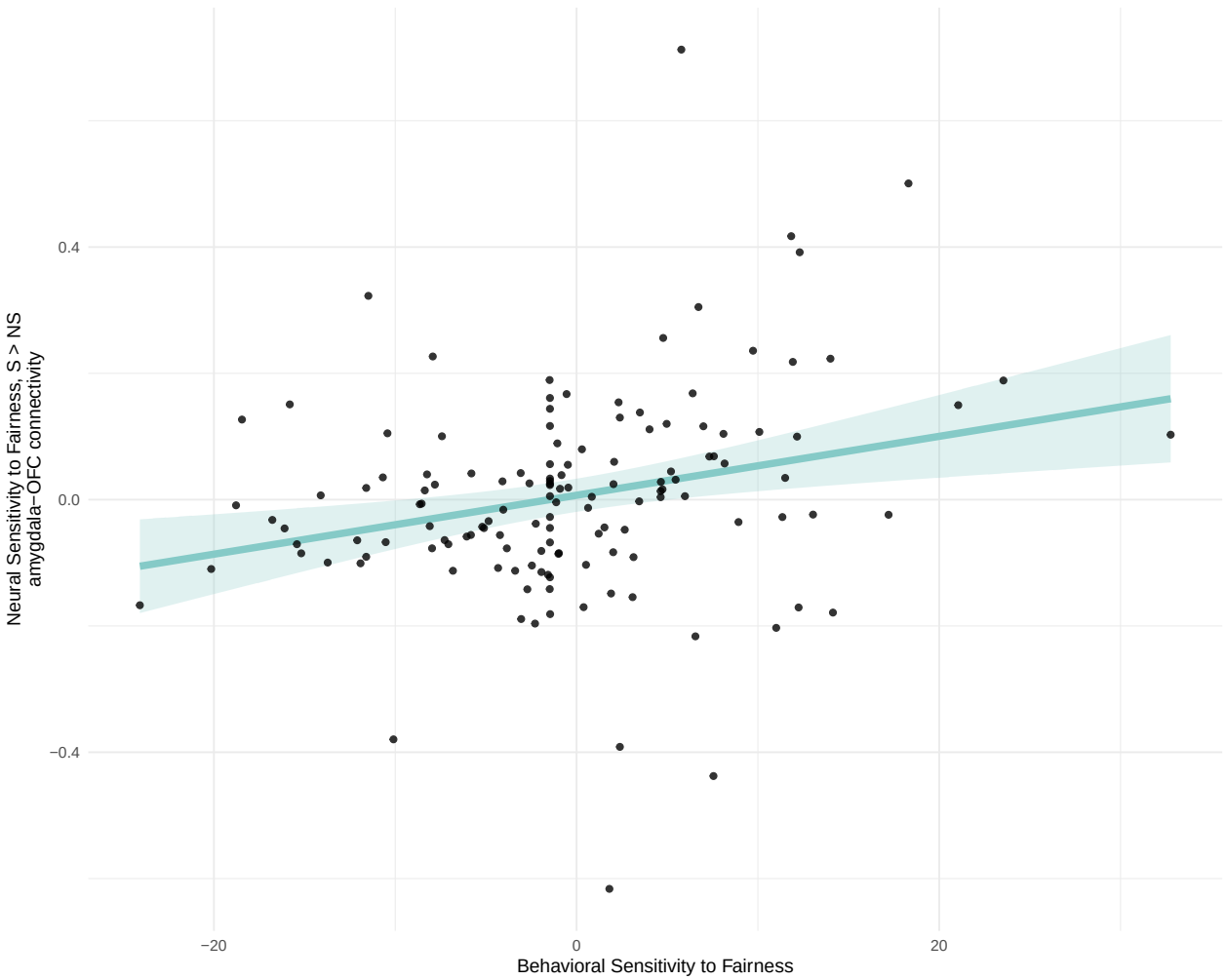
```
## -0.6162 -0.0815 -0.0045  0.0828  0.6627
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.02603    0.07355   -0.35     0.72
## sub_age_MC    -0.00164    0.00153   -1.07     0.29
## gender_female  0.02345    0.07535    0.31     0.76
## gender_male    0.04910    0.07836    0.63     0.53
## avg_fdmean_MC  0.15542    0.22784    0.68     0.50
## avg_tsnr_MC   -0.00183    0.00231   -0.79     0.43
##
## Residual standard error: 0.16 on 132 degrees of freedom
## Multiple R-squared:  0.027,  Adjusted R-squared:  -0.00982
## F-statistic: 0.733 on 5 and 132 DF,  p-value: 0.6
```

```r
diff_beh <- lm(diff_slope ~ sub_age_MC + gender_female + gender_male + avg_fdmean_MC + avg_tsnr_MC, data
# BEHAV COVAR OF INTEREST AS DV
summary(diff_beh)
```

```
##
## Call:
## lm(formula = diff_slope ~ sub_age_MC + gender_female + gender_male +
##     avg_fdmean_MC + avg_tsnr_MC, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -23.317  -4.806  -0.383   5.026  31.618
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.5386     4.0917   -0.38     0.71
## sub_age_MC      0.0712     0.0853    0.83     0.41
## gender_female   0.8020     4.1917    0.19     0.85
## gender_male     1.8358     4.3591    0.42     0.67
## avg_fdmean_MC  -0.2555    12.6753   -0.02     0.98
## avg_tsnr_MC    -0.1600     0.1288   -1.24     0.22
##
## Residual standard error: 9 on 132 degrees of freedom
## Multiple R-squared:  0.0336, Adjusted R-squared:  -0.00303
## F-statistic: 0.917 on 5 and 132 DF,  p-value: 0.472
```

```r
ggplot(df, aes(x = diff_slope, y = `amyg_AlphaDiff_C12`)) +
  geom_smooth(method = "lm", formula = y ~ x,
    se = TRUE, color = "#85CAC7", fill = "#85CAC7", size = 2, alpha = 0.25) +
    geom_point(color = "black", size = 1.5, alpha = 0.8) +
  labs(x = "Behavioral Sensitivity to Fairness",
       y = "Neural Sensitivity to Fairness, S > NS\namygdala-OFC connectivity") +
  theme_minimal() +
  theme(legend.position="top")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Amygdala-dmPFC, S > NS behavioral sensitivity, Extrinsic Worsening (Other-Directed Affect Worsening)

```
# amygdala
model_amyg <- lm(`amyg_AlphaDiffExtWor_C12` ~ sub_age_MC + gender_female + gender_male + avg_fdmean_MC +
summary(model_amyg)
```

```
##
## Call:
## lm(formula = amyg_AlphaDiffExtWor_C12 ~ sub_age_MC + gender_female +
##     gender_male + avg_fdmean_MC + avg_tsnr_MC + diff_slope +
##     avg_ext_wor.x + (diff_slope * avg_ext_wor.x), data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.23957 -0.05384  0.00557  0.05474  0.28506
##
## Coefficients:
##                          Estimate Std. Error t value  Pr(>|t|)
## (Intercept)             -0.048307   0.051753   -0.93      0.35
## sub_age_MC               0.000497   0.000917    0.54      0.59
```

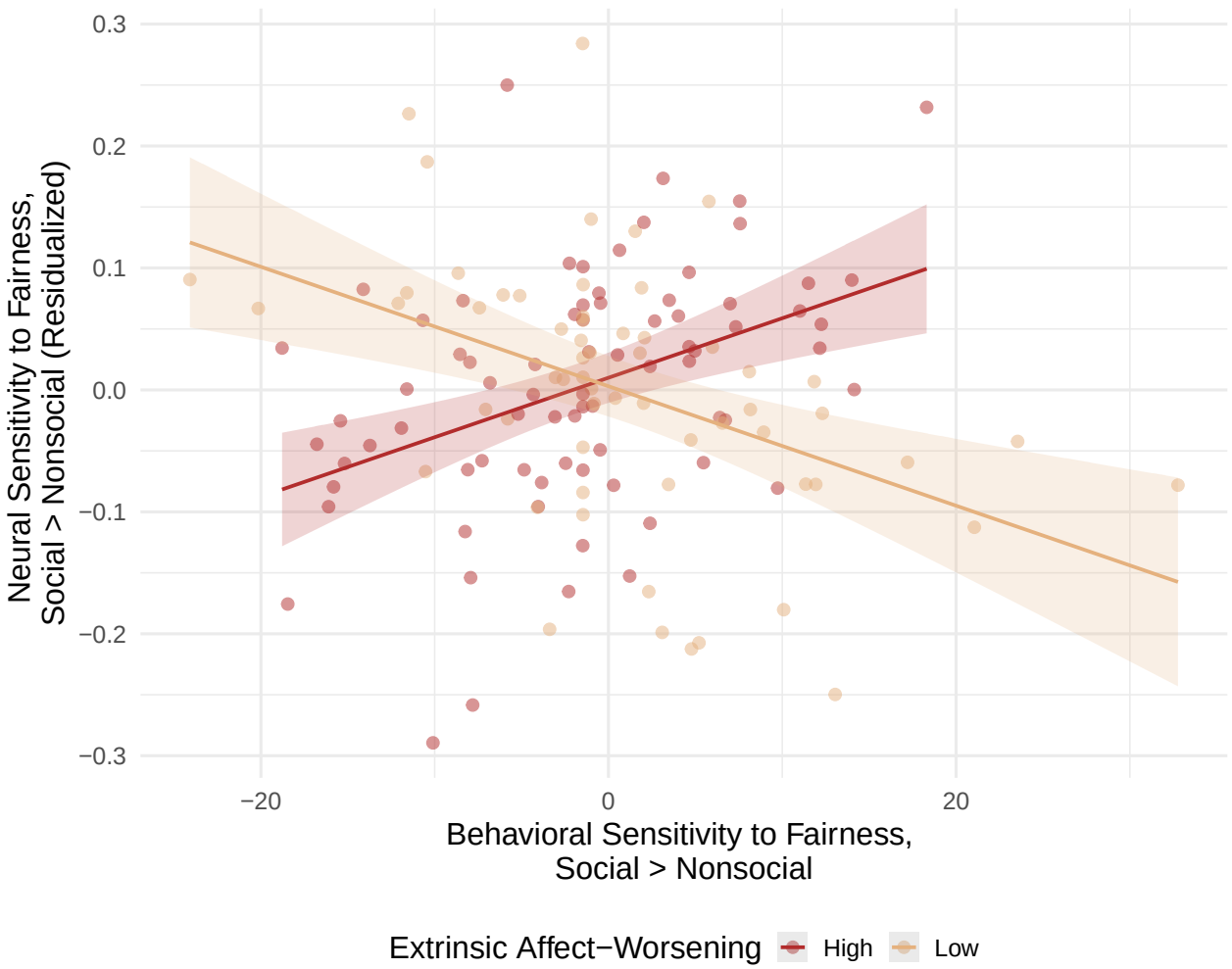
```
## gender_female             0.008836   0.044843    0.20      0.84
## gender_male               0.035407   0.046641    0.76      0.45
## avg_fdmean_MC             0.059723   0.135075    0.44      0.66
## avg_tsnr_MC               0.001796   0.001381    1.30      0.20
## diff_slope               -0.014396   0.003064   -4.70 0.0000066 ***
## avg_ext_wor.x             0.018358   0.017167    1.07      0.29
## diff_slope:avg_ext_wor.x  0.010813   0.002217    4.88 0.0000031 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.096 on 129 degrees of freedom
## Multiple R-squared:  0.167,  Adjusted R-squared:  0.115
## F-statistic: 3.23 on 8 and 129 DF,  p-value: 0.00223
```

```r
# Create High/Low grouping by median split
df$extwor_group <- ifelse(df$avg_ext_wor.x >= median(df$avg_ext_wor.x, na.rm = TRUE), "High", "Low")

cbextwor_pair <- SteppedSequential5Steps[c(2, 9)]
names(cbextwor_pair) <- c("High", "Low")

# Regression scatterplot with colorblind-safe palette and residualized Y axis
ggplot(df, aes(x = diff_slope,
               y = residuals(lm(`amyg_AlphaDiffExtWor_C12` ~ sub_age_MC + gender_female + gender_male +
               color = extwor_group,
               fill = extwor_group)) +
  geom_point(alpha = 0.5, size = 3) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.2) +
  scale_color_manual(values = cbextwor_pair) +
  scale_fill_manual(values = cbextwor_pair) +
  labs(x = "Behavioral Sensitivity to Fairness, \nSocial > Nonsocial",
       y = "Neural Sensitivity to Fairness, \nSocial > Nonsocial (Residualized)",
       color = "Extrinsic Affect-Worsening") +
  theme_minimal(base_size = 18) +
  guides(fill = "none", shape = "none") +
  theme(legend.position="bottom")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



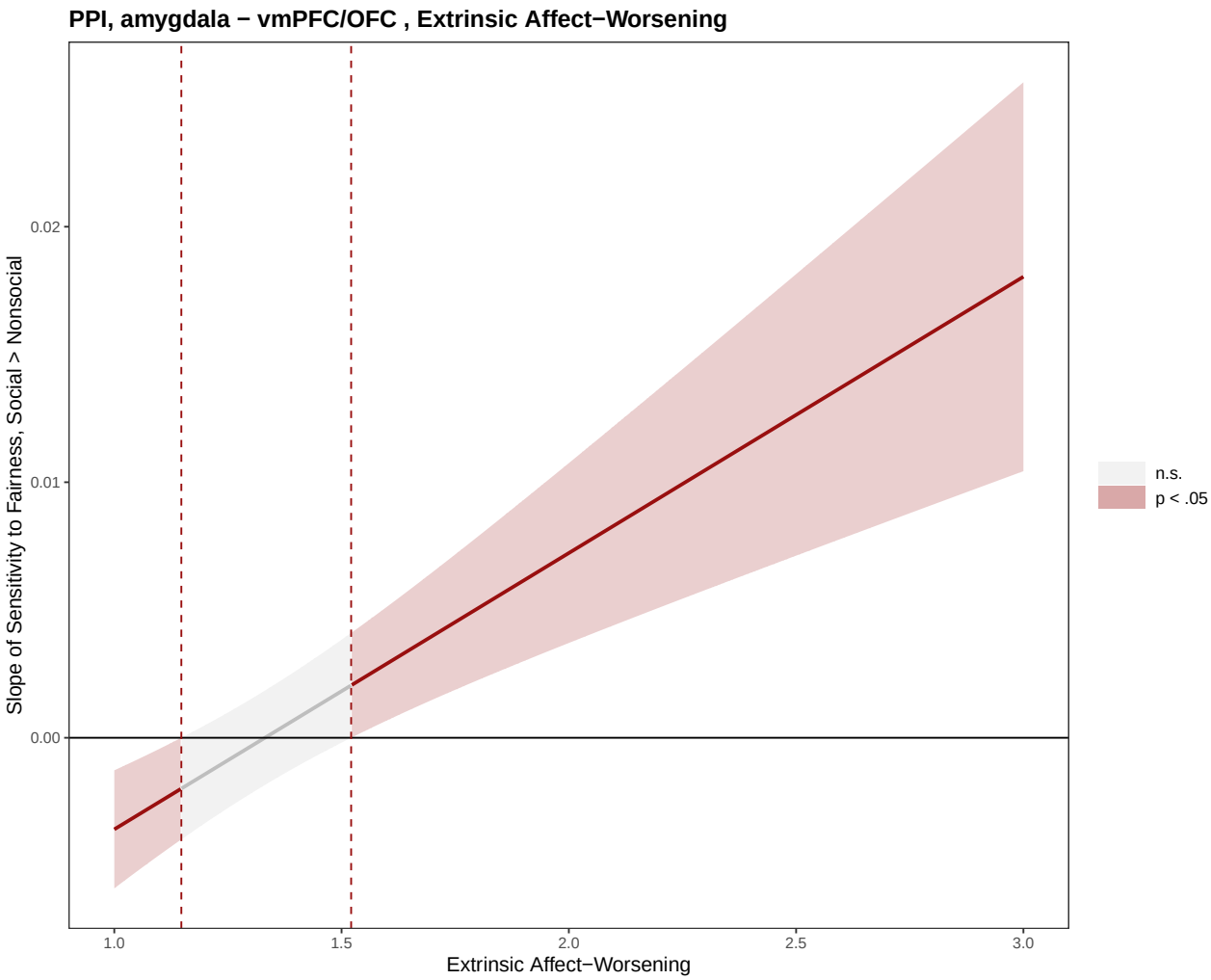

```
# Johnson-Neyman plot version
m_jn <- lm(`amyg_AlphaDiffExtWor_C12` ~ diff_slope * avg_ext_wor.x +  sub_age_MC + gender_female + gende

# Obs range of moderator
mf <- model.frame(m_jn)
mod_rng <- range(mf$avg_ext_wor.x, na.rm = TRUE)

# Johnson-Neyman analysis
jn <- johnson_neyman(m_jn,
  pred = "diff_slope",
  modx = "avg_ext_wor.x",
  alpha = 0.05,
  mod.range = mod_rng,
  sig.color = cb_palette[1],
  insig.color = "gray",
  line.thickness = 1,
  y.label = "Slope of Sensitivity to Fairness, Social > Nonsocial",
  modx.label = "Extrinsic Affect-Worsening",
  title = "PPI, amygdala - vmPFC/OFC , Extrinsic Affect-Worsening")
jn
```

```
## JOHNSON-NEYMAN INTERVAL
##
```

```
## When avg_ext_wor.x is OUTSIDE the interval [1.15, 1.52], the slope of
## diff_slope is p < .05.
##
## Note: The range of observed values of avg_ext_wor.x is [1.00, 3.00]
```



Behavioral Plots (supplement)

```r
ext_wor_all <- summary(lm(slope_all ~ avg_ext_wor.x, df))
ext_wor_all
```

```
##
## Call:
## lm(formula = slope_all ~ avg_ext_wor.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -23.35  -9.35   1.27   8.35  26.08
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      11.90       2.93    4.06 0.000082 ***
```

```
## avg_ext_wor.x     4.35       1.98    2.20    0.029 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11 on 136 degrees of freedom
## Multiple R-squared:  0.0344, Adjusted R-squared:  0.0273
## F-statistic: 4.84 on 1 and 136 DF,  p-value: 0.0295
```

```
ext_bet_all <- summary(lm(slope_all ~ avg_ext_bet.x, df))
ext_bet_all
```

```
##
## Call:
## lm(formula = slope_all ~ avg_ext_bet.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -24.18 -10.74   1.63   8.51  26.09
##
## Coefficients:
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept)      24.67       5.20    4.74 0.0000053 ***
## avg_ext_bet.x    -1.69       1.29   -1.31      0.19
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0125, Adjusted R-squared:  0.0052
## F-statistic: 1.72 on 1 and 136 DF,  p-value: 0.192
```

```
int_wor_all <- summary(lm(slope_all ~ avg_int_wor.x, df))
int_wor_all
```

```
##
## Call:
## lm(formula = slope_all ~ avg_int_wor.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -24.86 -10.13   1.51   8.64  27.58
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      11.73       3.71    3.17   0.0019 **
## avg_int_wor.x     2.41       1.38    1.75   0.0832 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11 on 136 degrees of freedom
## Multiple R-squared:  0.0219, Adjusted R-squared:  0.0147
## F-statistic: 3.05 on 1 and 136 DF,  p-value: 0.0832
```

```
int_bet_all <- summary(lm(slope_all ~ avg_int_bet.x, df))
int_bet_all
```

```
##
```

```
## Call:
## lm(formula = slope_all ~ avg_int_bet.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -25.09 -11.11   2.53   8.71  24.31
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)     18.103      4.718    3.84  0.00019 ***
## avg_int_bet.x   -0.044      1.590   -0.03  0.97797
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  5.63e-06,   Adjusted R-squared:  -0.00735
## F-statistic: 0.000765 on 1 and 136 DF,  p-value: 0.978
```

```r
ext_wor_soc <- summary(lm(slope_social ~ avg_ext_wor.x, df))
ext_wor_soc
```

```
##
## Call:
## lm(formula = slope_social ~ avg_ext_wor.x, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -28.433 -10.343   0.863  10.780  21.159
##
## Coefficients:
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept)      15.18       3.13    4.86 0.0000032 ***
## avg_ext_wor.x     2.86       2.11    1.36      0.18
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0133, Adjusted R-squared:  0.00607
## F-statistic: 1.84 on 1 and 136 DF,  p-value: 0.178
```

```r
ext_bet_soc <- summary(lm(slope_social ~ avg_ext_bet.x, df))
ext_bet_soc
```

```
##
## Call:
## lm(formula = slope_social ~ avg_ext_bet.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -28.68 -11.41   1.99  10.61  21.74
##
## Coefficients:
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept)      25.80       5.50    4.69 0.0000065 ***
## avg_ext_bet.x    -1.67       1.36   -1.23      0.22
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0109, Adjusted R-squared:  0.00364
## F-statistic:  1.5 on 1 and 136 DF,  p-value: 0.223
```

```r
int_wor_soc <- summary(lm(slope_social ~ avg_int_wor.x, df))
int_wor_soc
```

```
##
## Call:
## lm(formula = slope_social ~ avg_int_wor.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -29.35 -10.94   0.54   9.85  23.30
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      12.86       3.92    3.28   0.0013 **
## avg_int_wor.x     2.44       1.46    1.67   0.0965 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0202, Adjusted R-squared:  0.013
## F-statistic:  2.8 on 1 and 136 DF,  p-value: 0.0965
```

```r
int_bet_soc <- summary(lm(slope_social ~ avg_int_bet.x, df))
int_bet_soc
```

```
##
## Call:
## lm(formula = slope_social ~ avg_int_bet.x, data = df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -29.77 -11.75   1.56  11.12  20.67
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)     21.653      4.979    4.35 0.000027 ***
## avg_int_bet.x   -0.852      1.678   -0.51     0.61
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.00189,    Adjusted R-squared:  -0.00545
## F-statistic: 0.257 on 1 and 136 DF,  p-value: 0.613
```

```r
ext_wor_ns <- summary(lm(slope_nonsocial ~ avg_ext_wor.x, df))
ext_wor_ns
```

```
##
## Call:
```

```
## lm(formula = slope_nonsocial ~ avg_ext_wor.x, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -25.258 -10.807  -0.409  10.655  22.620
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      12.49       2.96    4.22 0.000045 ***
## avg_ext_wor.x     5.09       2.00    2.55    0.012 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0455, Adjusted R-squared:  0.0385
## F-statistic: 6.49 on 1 and 136 DF,  p-value: 0.012
```

```r
ext_bet_ns <- summary(lm(slope_nonsocial ~ avg_ext_bet.x, df))
ext_bet_ns
```

```
##
## Call:
## lm(formula = slope_nonsocial ~ avg_ext_bet.x, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -26.344 -10.643  -0.276  10.102  22.419
##
## Coefficients:
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept)      26.63       5.29    5.03 0.0000015 ***
## avg_ext_bet.x    -1.77       1.31   -1.35      0.18
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0132, Adjusted R-squared:  0.00597
## F-statistic: 1.82 on 1 and 136 DF,  p-value: 0.179
```

```r
int_wor_ns <- summary(lm(slope_nonsocial ~ avg_int_wor.x, df))
int_wor_ns
```

```
##
## Call:
## lm(formula = slope_nonsocial ~ avg_int_wor.x, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -27.098 -10.243  -0.585  10.467  23.352
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      14.28       3.78    3.77  0.00024 ***
## avg_int_wor.x     2.06       1.41    1.46  0.14644
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.0154, Adjusted R-squared:  0.0082
## F-statistic: 2.13 on 1 and 136 DF,  p-value: 0.146
```

```r
int_bet_ns <- summary(lm(slope_nonsocial ~ avg_int_bet.x, df))
int_bet_ns
```

```
##
## Call:
## lm(formula = slope_nonsocial ~ avg_int_bet.x, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -27.195 -10.644  -0.725  11.137  20.946
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)     18.468      4.800    3.85  0.00018 ***
## avg_int_bet.x    0.394      1.618    0.24  0.80797
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 136 degrees of freedom
## Multiple R-squared:  0.000436,   Adjusted R-squared:  -0.00691
## F-statistic: 0.0593 on 1 and 136 DF,  p-value: 0.808
```

```r
# Difference between social and nonsocial context, extrinsic worsening trait in behavior (no difference
df_long_beh <- data.frame(slope = c(df$slope_social, df$slope_nonsocial),
  eros  = c(df$avg_ext_wor.x, df$avg_ext_wor.x),
  context = factor(rep(c("social", "nonsocial"), each = nrow(df))))

model_contextdiff <- lm(slope ~ eros * context, df_long_beh)
summary(model_contextdiff)
```

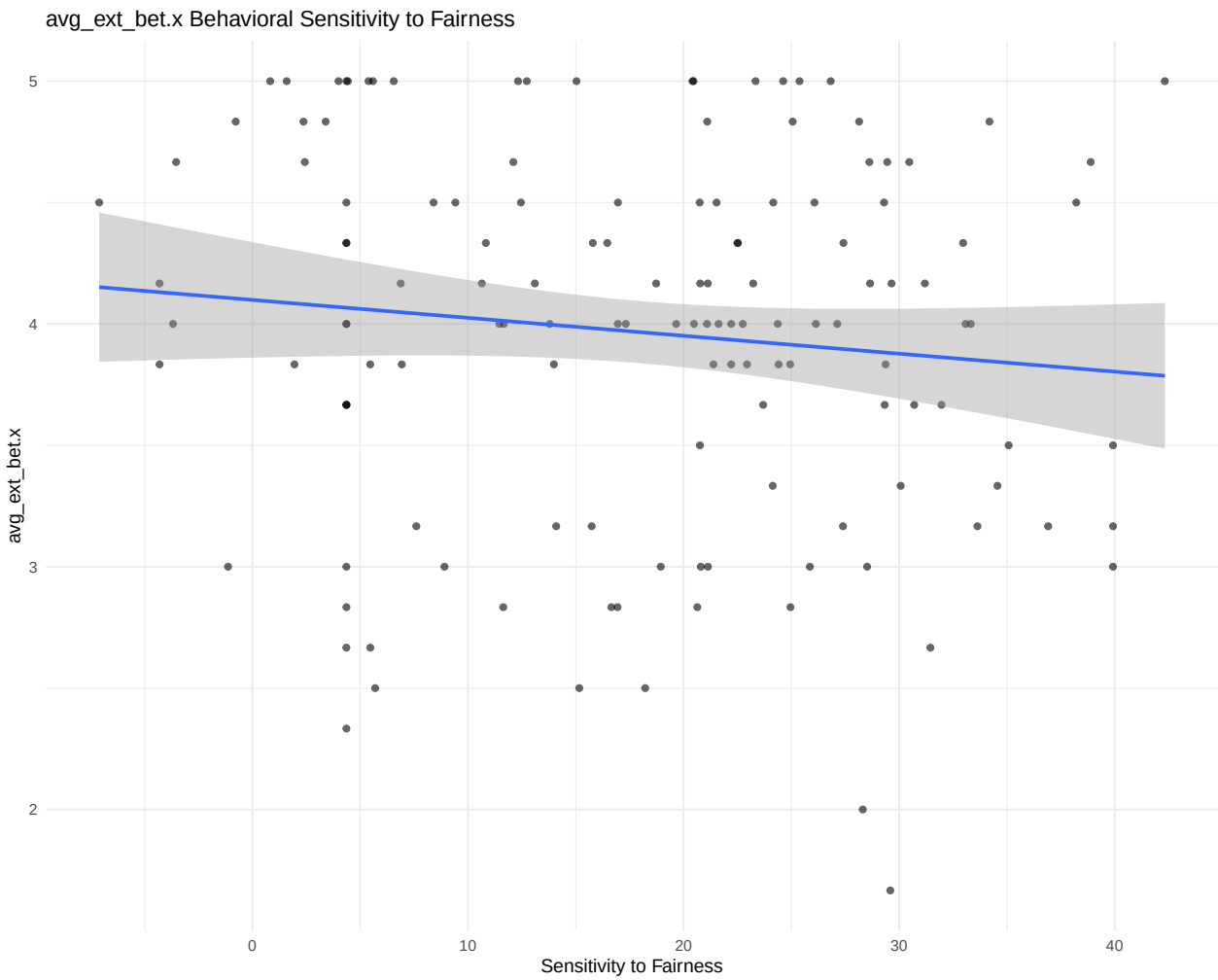
```
##
## Call:
## lm(formula = slope ~ eros * context, data = df_long_beh)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -28.433 -10.749   0.118  10.753  22.620
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)           12.49       3.05    4.10 0.000054 ***
## eros                   5.09       2.06    2.48    0.014 *
## contextsocial          2.69       4.31    0.63    0.532
## eros:contextsocial    -2.23       2.91   -0.77    0.443
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12 on 272 degrees of freedom
## Multiple R-squared:  0.0291, Adjusted R-squared:  0.0184
```

```
## F-statistic: 2.72 on 3 and 272 DF,  p-value: 0.0448
```

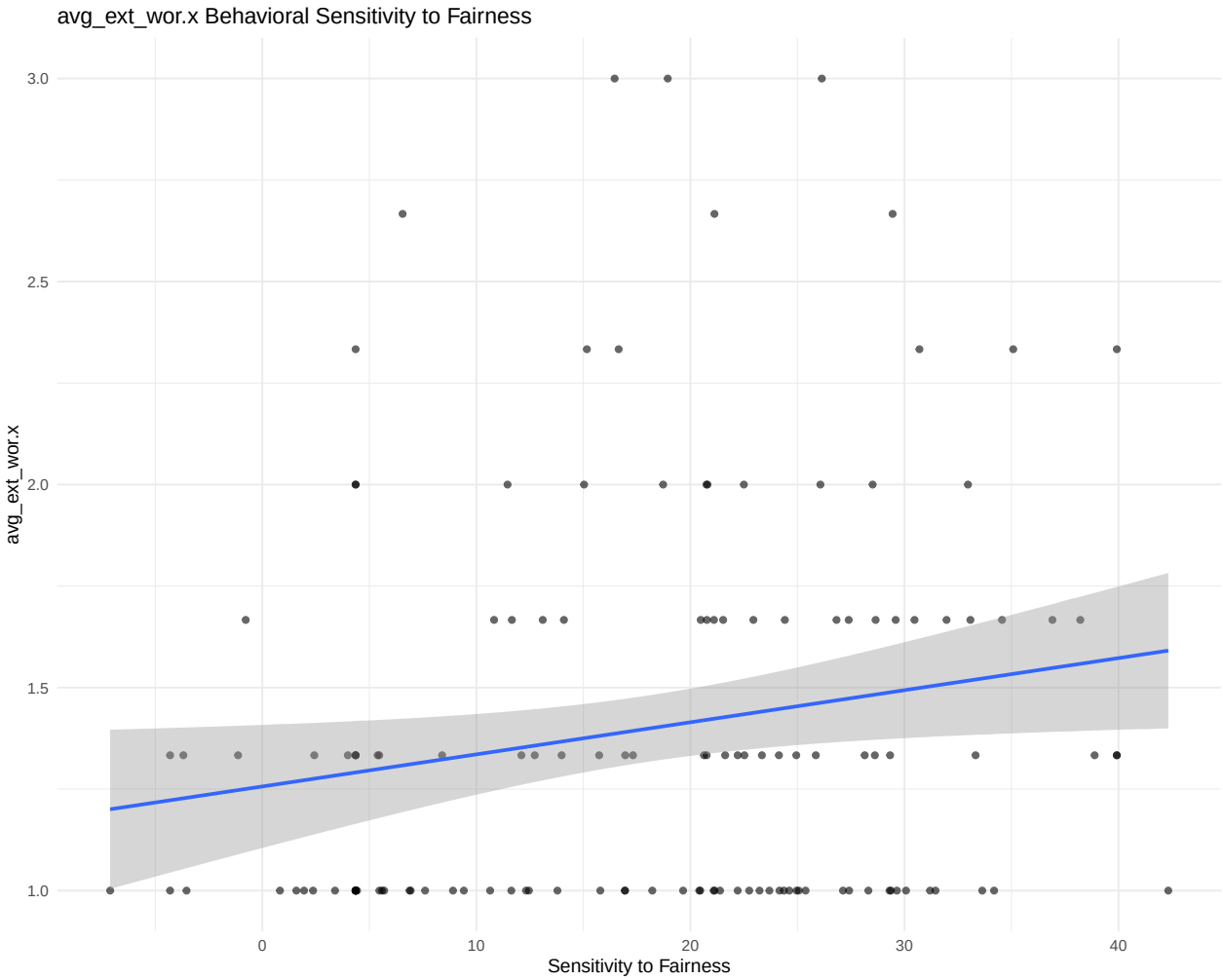
```r
vars <- c("avg_ext_bet.x", "avg_ext_wor.x","avg_int_bet.x", "avg_int_wor.x")

for (v in vars) {
  p <- ggplot(df, aes(x = slope_all, y = .data[[v]])) +
    geom_point(alpha = 0.6) +
    geom_smooth(method = "lm", se = TRUE) +
    theme_minimal() +
    labs(title = paste(v, "Behavioral Sensitivity to Fairness"),
      x = "Sensitivity to Fairness",
      y = v)
    print(p)}
```

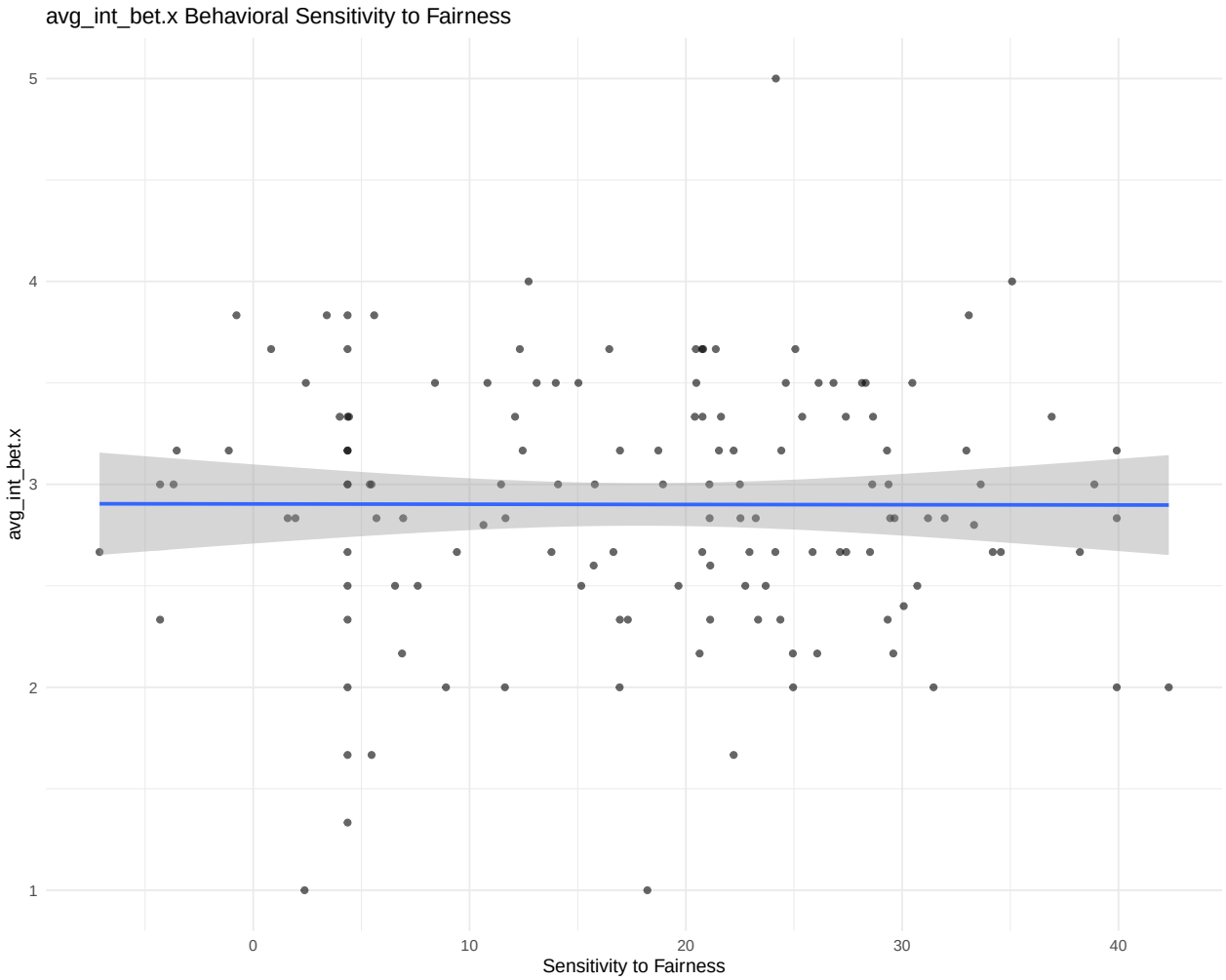
```
## `geom_smooth()` using formula = 'y ~ x'
```



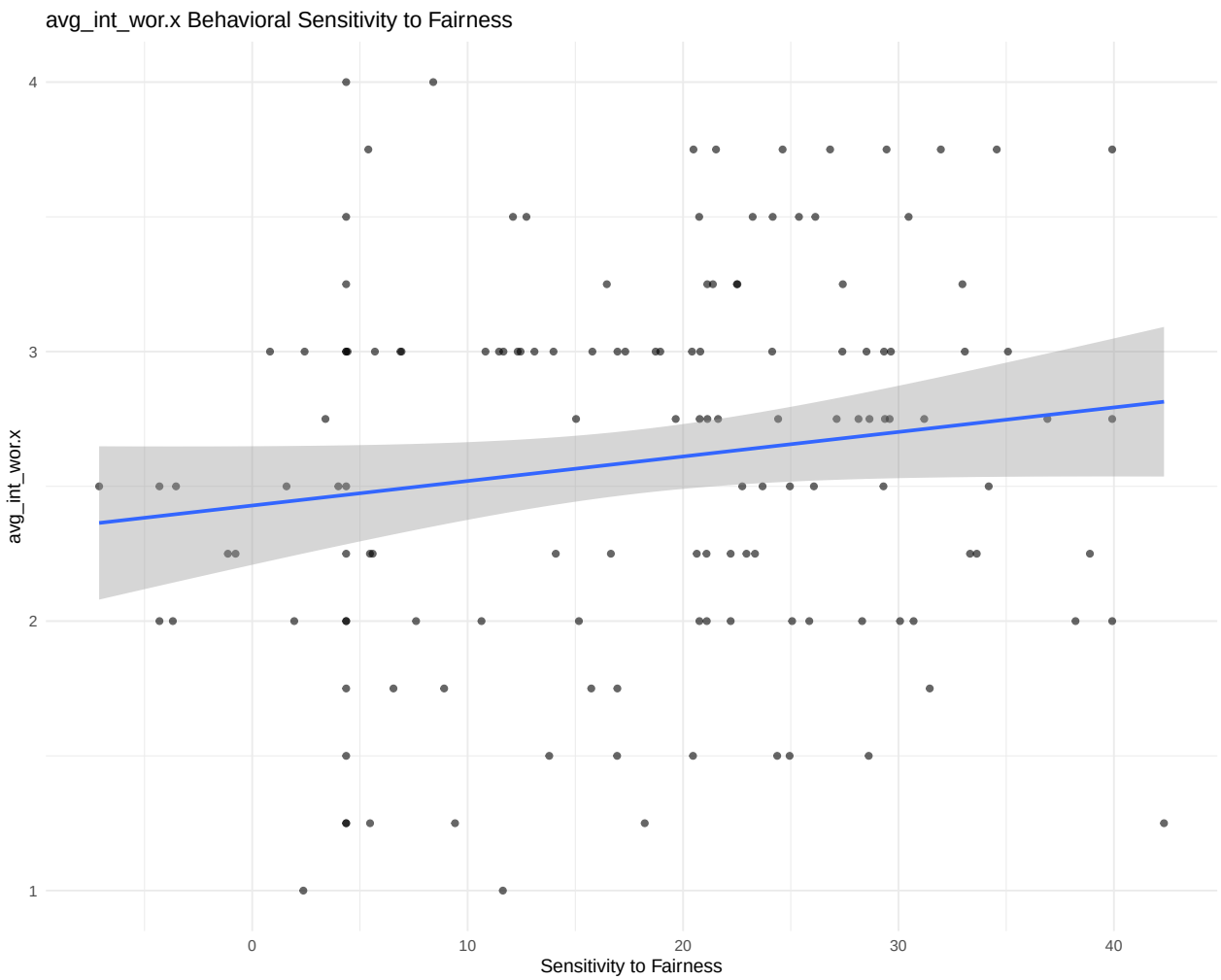

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```



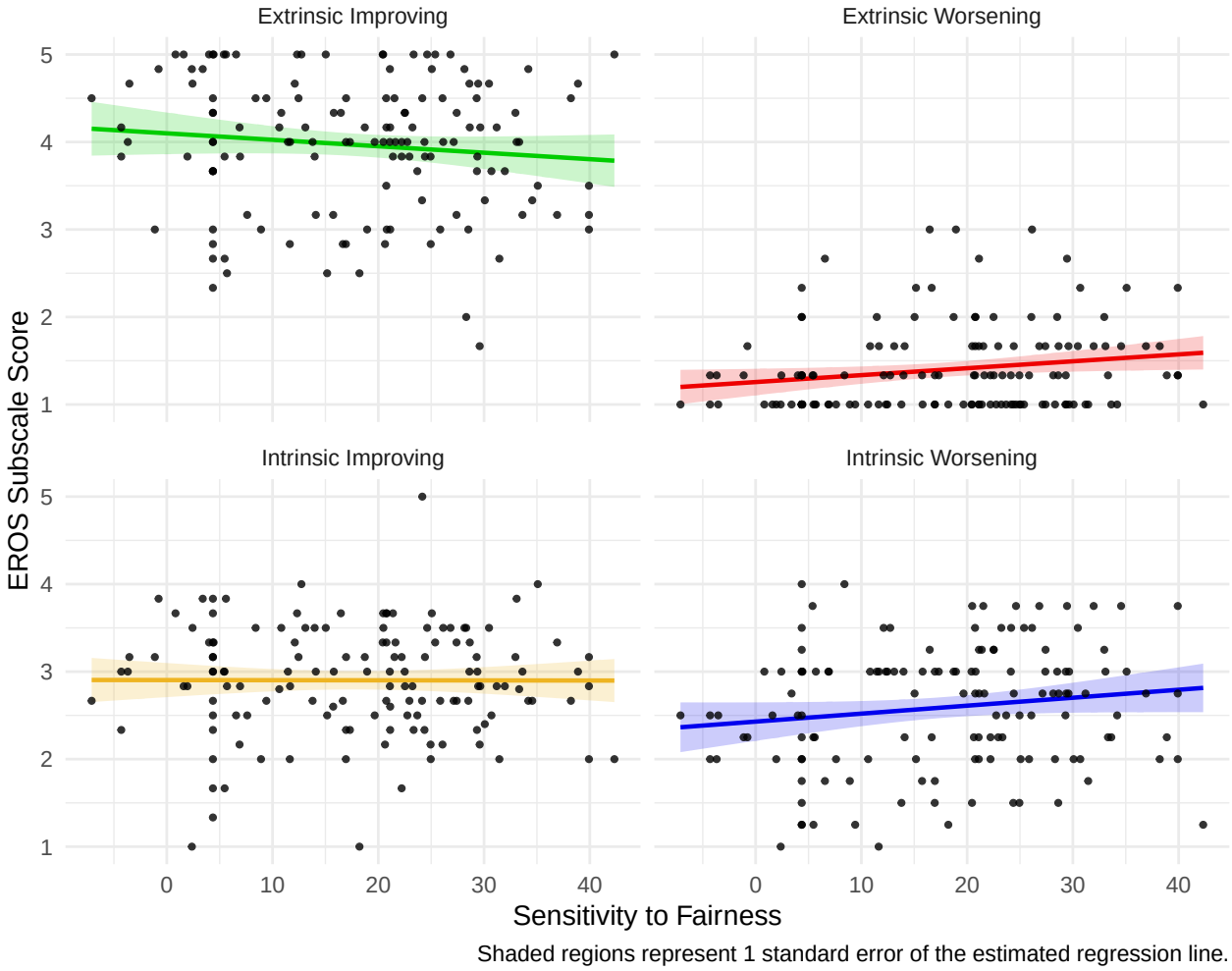

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
df_long <- df |>
  select(slope_all, avg_ext_bet.x, avg_ext_wor.x, avg_int_bet.x, avg_int_wor.x) |>
  pivot_longer(-slope_all, names_to = "EROS_scale", values_to = "score")

df_long$EROS_scale <- factor(df_long$EROS_scale,
  levels = c("avg_ext_bet.x", "avg_ext_wor.x", "avg_int_bet.x", "avg_int_wor.x"),
  labels = c("Extrinsic Improving", "Extrinsic Worsening", "Intrinsic Improving", "Intrinsic Worsening"

ggplot(df_long, aes(x = slope_all, y = score)) +
  geom_smooth(aes(color = EROS_scale, fill = EROS_scale), method = "lm", se = TRUE, linewidth = 1.2, al
  geom_point(color = "black", alpha = 0.8) +
  facet_wrap(~ EROS_scale) +
  scale_color_manual(values = c("Extrinsic Improving" = "green3",
                                "Extrinsic Worsening" = "red2",
                                "Intrinsic Improving" = "goldenrod2",
                                "Intrinsic Worsening" = "blue2")) +
  scale_fill_manual(values = c("Extrinsic Improving" = "green3",
                               "Extrinsic Worsening" = "red2",
                               "Intrinsic Improving" = "goldenrod2",
                               "Intrinsic Worsening" = "blue2")) +
  guides(color = "none", fill = "none") +
  labs(x = "Sensitivity to Fairness",
```

```r
       y = "EROS Subscale Score",
       caption = "Shaded regions represent 1 standard error of the estimated regression line.") +
  theme_minimal(base_size = 16)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```